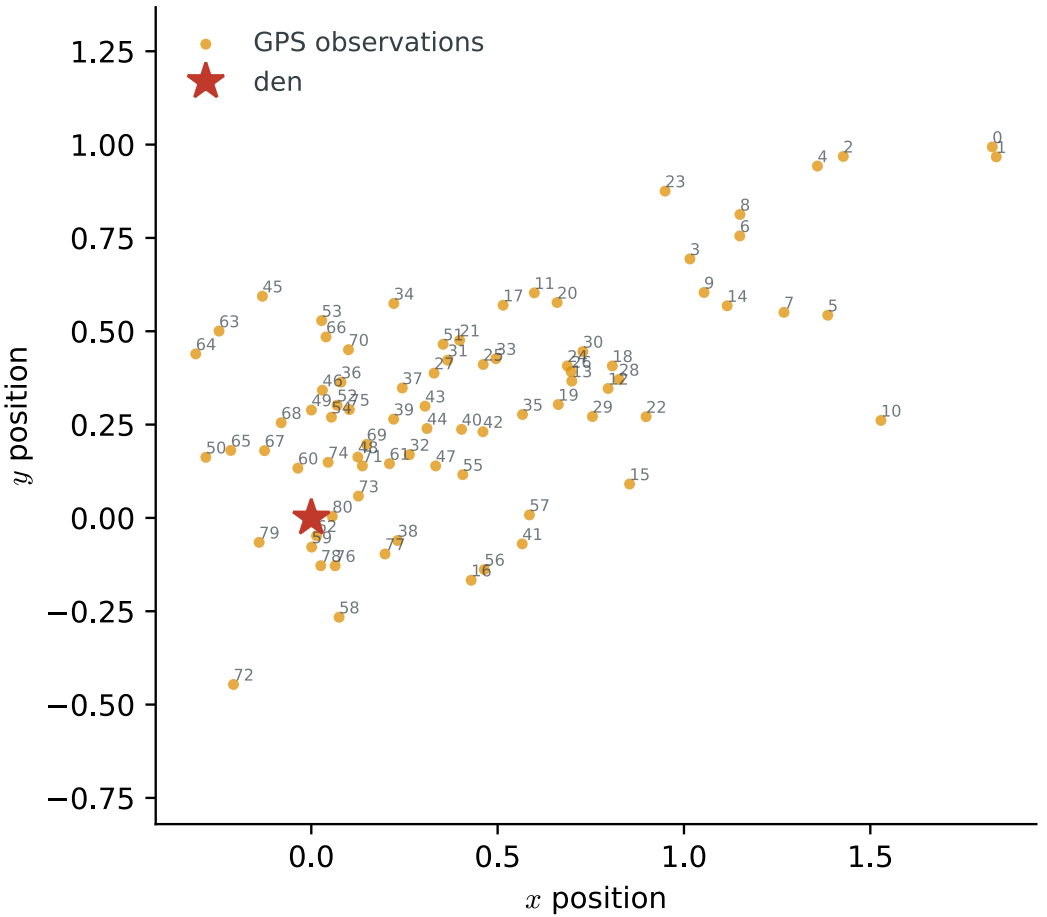
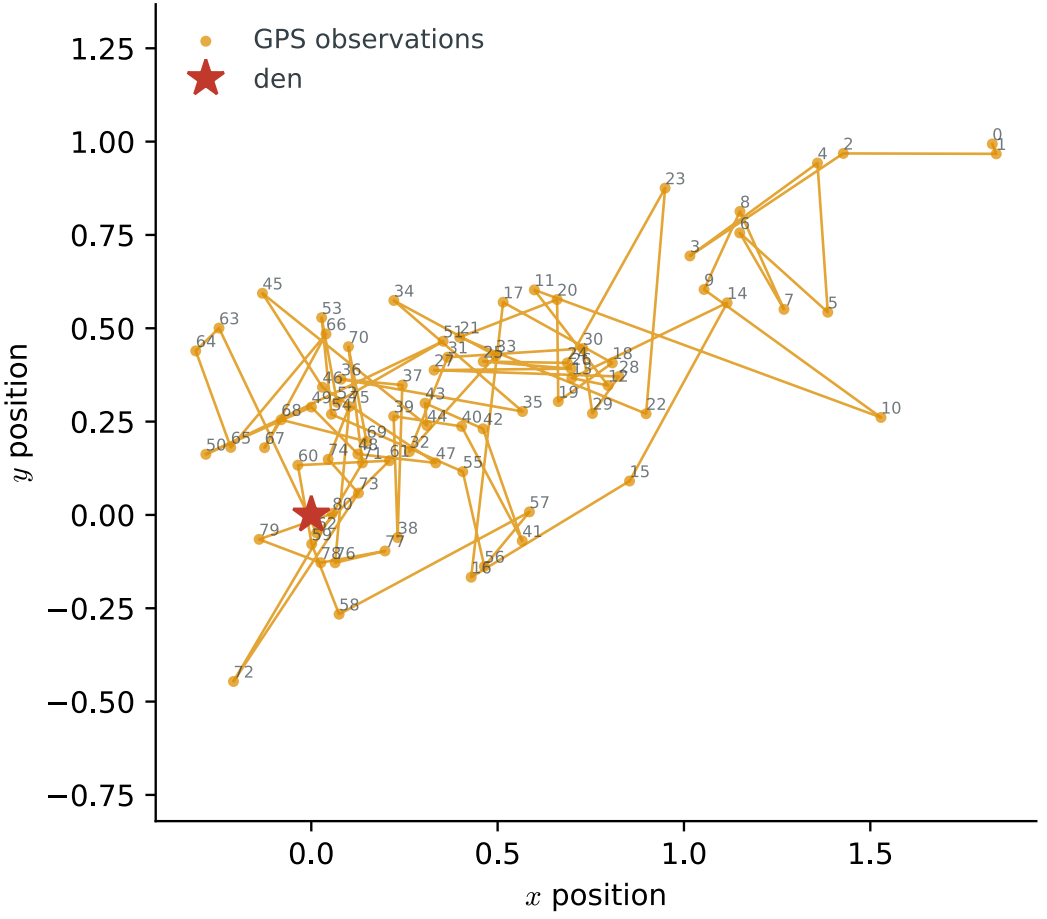
Foundations of Data Science

Lecture 5: State-space models

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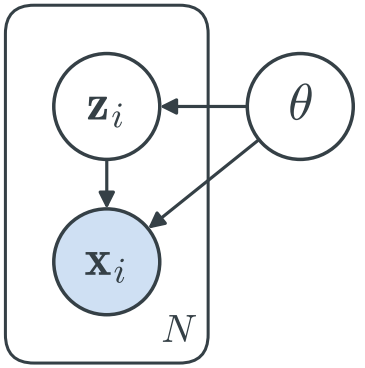


Today's case study: **tracking** the location of a wolf over time from noisy GPS observations, here simulated.



The same observations, joined in the order they were recorded.

Discrete-time models



Static latent variable models

A latent variable model explains N observations $\mathbf{x}_i$ with local latent variables $\mathbf{z}_i$ and global parameters θ,

$$p(\mathbf{x}_{1:N}, \mathbf{z}_{1:N}, \theta) = p(\theta) \prod_{i=1}^{N} p(\mathbf{x}_i|\mathbf{z}_i, \theta) p(\mathbf{z}_i|\theta).$$

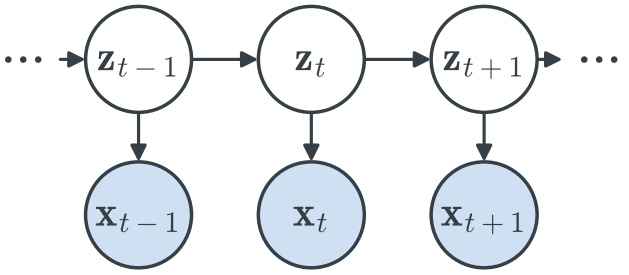


What if the system evolves over time and we have a sequence of observations $\mathbf{x}_{1:T} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)$ collected at discrete time steps $t = 1, \ldots, T$?

State-space models

To model a dynamical system, we can first assume a discretization of time and introduce a sequence of latent variables $\mathbf{z}_t$ that represent the state of the system at each time step t.

In this context, a **state-space model** is a latent variable model that explains a sequence $\mathbf{x}_{1:T} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)$ of observations in terms of a sequence $\mathbf{z}_{1:T} = (\mathbf{z}_1, \ldots, \mathbf{z}_T)$ of latent variables. Each latent variable now depends on the previous one: the plate becomes a chain.



In a Markovian state-space model, the latent variables form a Markov chain

$$p(\mathbf{z}_t|\mathbf{z}_{1:t-1}) = p(\mathbf{z}_t|\mathbf{z}_{t-1}),$$

where the conditional distribution $p(\mathbf{z}_t|\mathbf{z}_{t-1})$ is called the **transition model**.

The observations are conditionally independent given the latent variables,

$$p(\mathbf{x}_t|\mathbf{x}_{1:t-1}, \mathbf{z}_{1:t}) = p(\mathbf{x}_t|\mathbf{z}_t),$$

where the conditional distribution $p(\mathbf{x}_t|\mathbf{z}_t)$ is called the **observation model**.



Example. We want to track the location of a wild animal (e.g., a wolf) over time using noisy GPS observations.

Assumptions:

- The animal has a home location (den, nest) at $\boldsymbol{\mu} \in \mathbb{R}^2$.
- The animal moves according to a random walk with drift towards the home location.
- The GPS observations are noisy measurements of the animal's true location.
- Time is discretized regularly every Δt time units.

State-space model:

- States $\mathbf{z}_t \in \mathbb{R}^2$ represent the true location of the animal at time t. They evolve as a random walk with drift towards the home location $\boldsymbol{\mu}$.

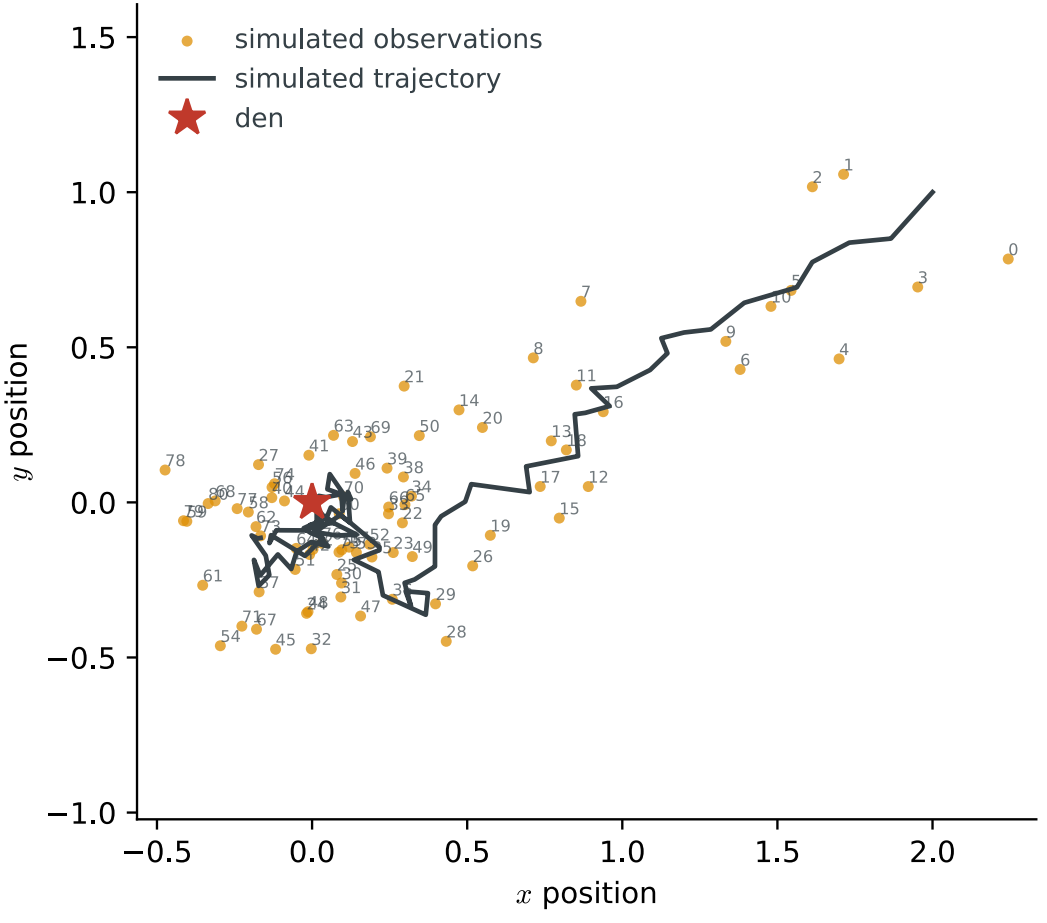
$$p(\mathbf{z}_t|\mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t|\mathbf{z}_{t-1} - \kappa(\mathbf{z}_{t-1} - \boldsymbol{\mu})\Delta t, \sigma^2 \Delta t \mathbf{I}),$$

  where $\kappa > 0$ is the strength of attraction to the home location and σ^2 is the variance of the random walk.

- Observations $\mathbf{x}_t \in \mathbb{R}^2$ represent noisy GPS measurements of the animal's location at time t. They relate to the states via

$$p(\mathbf{x}_t|\mathbf{z}_t) = \mathcal{N}(\mathbf{x}_t|\mathbf{z}_t, \mathbf{R}),$$

  where $\mathbf{R}$ is the observation noise covariance.

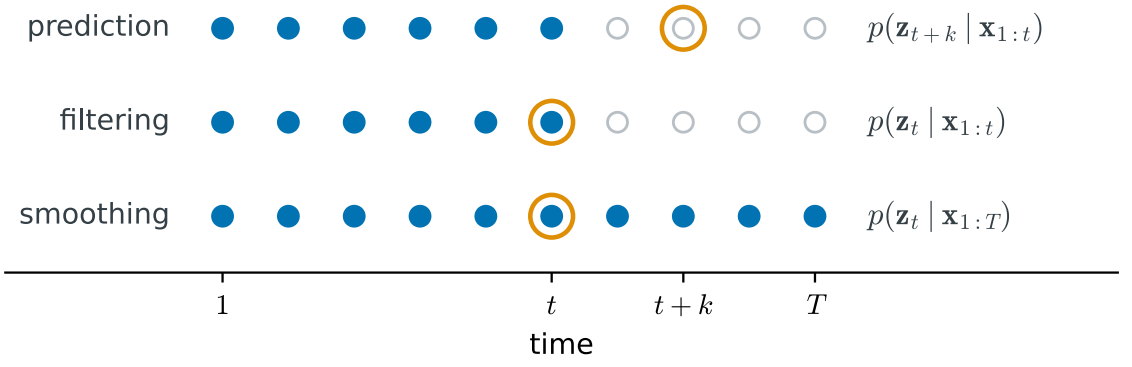


Example of trajectory and observations
generated from the discrete-time state-space model ($\Delta t = 0.25$).

Inference in state-space models

Given a state-space model and a sequence of observations $\mathbf{x}_{1:T}$, we are typically interested in solving one or more of the following inference problems:

- Prediction: $p(\mathbf{z}_{t+k}|\mathbf{x}_{1:t})$ for $k \geq 1$.
- Filtering: $p(\mathbf{z}_t|\mathbf{x}_{1:t})$.
- Smoothing: $p(\mathbf{z}_t|\mathbf{x}_{1:T})$.



Bayes filter

The Bayes filter is a recursive algorithm that computes the filtering distributions $p(\mathbf{z}_t|\mathbf{x}_{1:t})$ as

$$p(\mathbf{z}_t|\mathbf{x}_{1:t}) = \frac{p(\mathbf{x}_t|\mathbf{z}_t)\int p(\mathbf{z}_t|\mathbf{z}_{t-1})p(\mathbf{z}_{t-1}|\mathbf{x}_{1:t-1})d\mathbf{z}_{t-1}}{p(\mathbf{x}_t|\mathbf{x}_{1:t-1})},$$

for $t = 2, \ldots, T$, with the base case $p(\mathbf{z}_1|\mathbf{x}_1) = \frac{p(\mathbf{x}_1|\mathbf{z}_1)p(\mathbf{z}_1)}{p(\mathbf{x}_1)}$.

Proof. The filtering distribution can be derived in two steps:

1. Prediction: Push the filtering distribution from the previous time step through the transition model to obtain a prediction of the current state. That is,

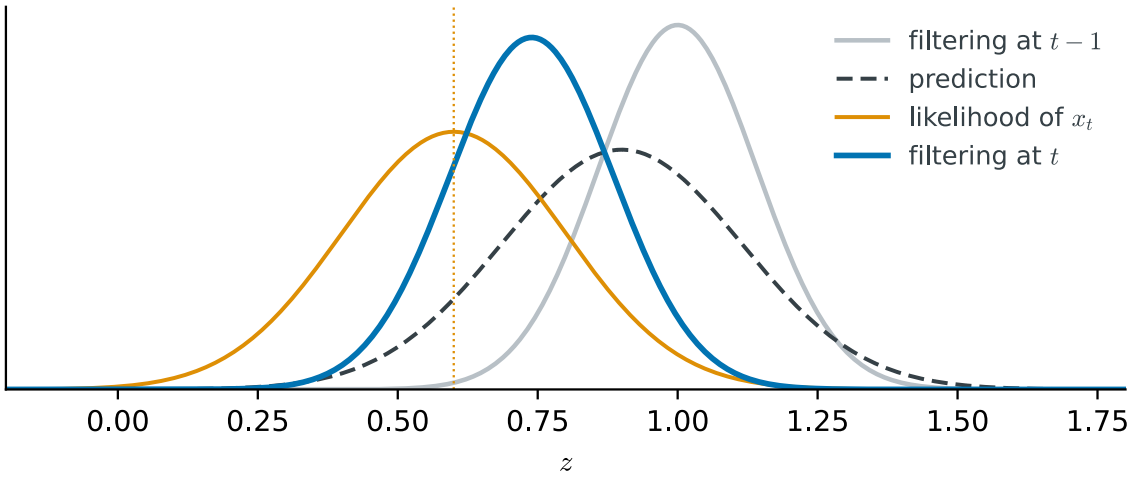
$$p(\mathbf{z}_t|\mathbf{x}_{1:t-1}) = \int p(\mathbf{z}_t|\mathbf{z}_{t-1})p(\mathbf{z}_{t-1}|\mathbf{x}_{1:t-1})d\mathbf{z}_{t-1}.$$

2. Update: Condition on the new observation to obtain the filtering distribution,

$$p(\mathbf{z}_t|\mathbf{x}_{1:t}) = \frac{p(\mathbf{x}_t|\mathbf{z}_t)p(\mathbf{z}_t|\mathbf{x}_{1:t-1})}{p(\mathbf{x}_t|\mathbf{x}_{1:t-1})},$$

   where the marginal likelihood $p(\mathbf{x}_t|\mathbf{x}_{1:t-1})$ is given by

$$p(\mathbf{x}_t|\mathbf{x}_{1:t-1}) = \int p(\mathbf{x}_t|\mathbf{z}_t)p(\mathbf{z}_t|\mathbf{x}_{1:t-1})d\mathbf{z}_t.$$



One step of the Bayes filter, in one dimension. The prediction spreads the previous filtering distribution; the update pulls it towards the new observation.

Once we have computed the filtering distributions $p(\mathbf{z}_t|\mathbf{x}_{1:t})$ for $t = 1, \ldots, T$, we can compute the **prediction distributions** $p(\mathbf{z}_{t+k}|\mathbf{x}_{1:t})$ for $k \geq 1$ using the prediction step of the Bayes filter iteratively,

$$p(\mathbf{z}_{t+k}|\mathbf{x}_{1:t}) = \int p(\mathbf{z}_{t+k}|\mathbf{z}_{t+k-1})p(\mathbf{z}_{t+k-1}|\mathbf{x}_{1:t})d\mathbf{z}_{t+k-1},$$

for $k = 1, 2, \ldots$.

Bayes smoother

The Bayes smoother computes the smoothing distributions $p(\mathbf{z}_t|\mathbf{x}_{1:T})$ for $t = 1, \ldots, T$ using the filtering distributions $p(\mathbf{z}_t|\mathbf{x}_{1:t})$ and the transition model $p(\mathbf{z}_t|\mathbf{z}_{t-1})$. It consists of a backward recursion,

$$p(\mathbf{z}_t|\mathbf{x}_{1:T}) = p(\mathbf{z}_t|\mathbf{x}_{1:t}) \int \frac{p(\mathbf{z}_{t+1}|\mathbf{z}_t)p(\mathbf{z}_{t+1}|\mathbf{x}_{1:T})}{p(\mathbf{z}_{t+1}|\mathbf{x}_{1:t})} d\mathbf{z}_{t+1},$$

for $t = T-1, T-2, \ldots, 1$, starting from $p(\mathbf{z}_T|\mathbf{x}_{1:T})$, the filtering distribution at $t = T$.

Proof. The joint distribution $p(\mathbf{z}_t, \mathbf{z}_{t+1}|\mathbf{x}_{1:T})$ can be computed as

$$\begin{aligned} p(\mathbf{z}_t, \mathbf{z}_{t+1}|\mathbf{x}_{1:T}) &= p(\mathbf{z}_t|\mathbf{z}_{t+1}, \mathbf{x}_{1:T})p(\mathbf{z}_{t+1}|\mathbf{x}_{1:T}) \\ &= p(\mathbf{z}_t|\mathbf{z}_{t+1}, \mathbf{x}_{1:t})p(\mathbf{z}_{t+1}|\mathbf{x}_{1:T}) \\ &= \frac{p(\mathbf{z}_{t+1}|\mathbf{z}_t)p(\mathbf{z}_t|\mathbf{x}_{1:t})}{p(\mathbf{z}_{t+1}|\mathbf{x}_{1:t})}p(\mathbf{z}_{t+1}|\mathbf{x}_{1:T}), \end{aligned}$$

where we used the conditional independence properties of the state-space model.

Marginalizing over $\mathbf{z}_{t+1}$ gives the desired result,

$$\begin{aligned} p(\mathbf{z}_t|\mathbf{x}_{1:T}) &= \int p(\mathbf{z}_t, \mathbf{z}_{t+1}|\mathbf{x}_{1:T})d\mathbf{z}_{t+1} \\ &= p(\mathbf{z}_t|\mathbf{x}_{1:t}) \int \frac{p(\mathbf{z}_{t+1}|\mathbf{z}_t)p(\mathbf{z}_{t+1}|\mathbf{x}_{1:T})}{p(\mathbf{z}_{t+1}|\mathbf{x}_{1:t})}d\mathbf{z}_{t+1}. \end{aligned}$$



> In general, the integrals of the Bayes filter and smoother have no closed form.

Two classes of models are exceptions:

- linear Gaussian models, where the integrals are Gaussian (Kalman filter and smoother);
- hidden Markov models, where they are finite sums (forward-backward algorithm).

Linear Gaussian state-space models

A linear Gaussian state-space model (LGSSM) is a state-space model where both the transition and observation models are linear Gaussian. That is,

$$p(\mathbf{z}_t|\mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t|\mathbf{A}\mathbf{z}_{t-1} + \mathbf{b}, \mathbf{Q}),$$
$$p(\mathbf{x}_t|\mathbf{z}_t) = \mathcal{N}(\mathbf{x}_t|\mathbf{H}\mathbf{z}_t, \mathbf{R}),$$

where $\mathbf{A}$ is the state transition matrix, $\mathbf{b}$ an offset, $\mathbf{Q}$ the process noise covariance, $\mathbf{H}$ the observation matrix, and $\mathbf{R}$ the observation noise covariance.

If the prior distribution $p(\mathbf{z}_1)$ is also Gaussian, then all filtering, prediction, and smoothing distributions are Gaussian.

> The wolf model is the case
>
> $$\mathbf{A} = (1 - \kappa\Delta t)\mathbf{I}, \quad \mathbf{b} = \kappa\Delta t\, \boldsymbol{\mu}, \quad \mathbf{Q} = \sigma^2\Delta t\, \mathbf{I}, \quad \mathbf{H} = \mathbf{I}.$$

The **Kalman filter** provides a closed-form expression for the filtering distributions in linear Gaussian state-space models.

At each time step t, $p(\mathbf{z}_t|\mathbf{x}_{1:t}) = \mathcal{N}(\mathbf{z}_t|\mathbf{m}_t, \mathbf{P}_t)$ is Gaussian with mean $\mathbf{m}_t$ and covariance $\mathbf{P}_t$. These parameters can be computed recursively by adapting the Bayes filter equations to the linear Gaussian case.

Proof. By recursion, assume that at time step $t-1$, the filtering distribution is Gaussian, $p(\mathbf{z}_{t-1}|\mathbf{x}_{1:t-1}) = \mathcal{N}(\mathbf{z}_{t-1}|\mathbf{m}_{t-1}, \mathbf{P}_{t-1})$, with the base case $p(\mathbf{z}_1|\mathbf{x}_1) = \mathcal{N}(\mathbf{z}_1|\mathbf{m}_1, \mathbf{P}_1)$.

For the prediction step, we have

$$
\begin{aligned}
p(\mathbf{z}_t|\mathbf{x}_{1:t-1}) &= \int p(\mathbf{z}_{t-1}|\mathbf{x}_{1:t-1}) p(\mathbf{z}_t|\mathbf{z}_{t-1}) d\mathbf{z}_{t-1} \\
&= \int \mathcal{N}(\mathbf{z}_{t-1}|\mathbf{m}_{t-1}, \mathbf{P}_{t-1}) \mathcal{N}(\mathbf{z}_t|\mathbf{A}\mathbf{z}_{t-1} + \mathbf{b}, \mathbf{Q}) d\mathbf{z}_{t-1} \\
&= \int \mathcal{N}\left(\begin{pmatrix} \mathbf{z}_{t-1} \\ \mathbf{z}_t \end{pmatrix} | \begin{bmatrix} \mathbf{m}_{t-1} \\ \mathbf{A}\mathbf{m}_{t-1} + \mathbf{b} \end{bmatrix}, \begin{bmatrix} \mathbf{P}_{t-1} & \mathbf{P}_{t-1}\mathbf{A}^T \\ \mathbf{A}\mathbf{P}_{t-1} & \mathbf{A}\mathbf{P}_{t-1}\mathbf{A}^T + \mathbf{Q} \end{bmatrix} \right) d\mathbf{z}_{t-1} \\
&= \mathcal{N}(\mathbf{z}_t|\mathbf{m}_t^-, \mathbf{P}_t^-),
\end{aligned}
$$

where $\mathbf{m}_t^- = \mathbf{A}\mathbf{m}_{t-1} + \mathbf{b}$ and $\mathbf{P}_t^- = \mathbf{A}\mathbf{P}_{t-1}\mathbf{A}^T + \mathbf{Q}$.

For the update step, we join the prediction distribution with the observation model,

$$p\left(\begin{pmatrix}\mathbf{z}_t\\ \mathbf{x}_t\end{pmatrix}|\mathbf{x}_{1:t-1}\right) = p(\mathbf{z}_t|\mathbf{x}_{1:t-1})p(\mathbf{x}_t|\mathbf{z}_t)$$
$$= \mathcal{N}\left(\begin{pmatrix}\mathbf{z}_t\\ \mathbf{x}_t\end{pmatrix} | \begin{bmatrix}\mathbf{m}_t^-\\ \mathbf{H}\mathbf{m}_t^-\end{bmatrix}, \begin{bmatrix}\mathbf{P}_t^- & \mathbf{P}_t^-\mathbf{H}^T\\ \mathbf{H}\mathbf{P}_t^- & \mathbf{H}\mathbf{P}_t^-\mathbf{H}^T + \mathbf{R}\end{bmatrix}\right).$$

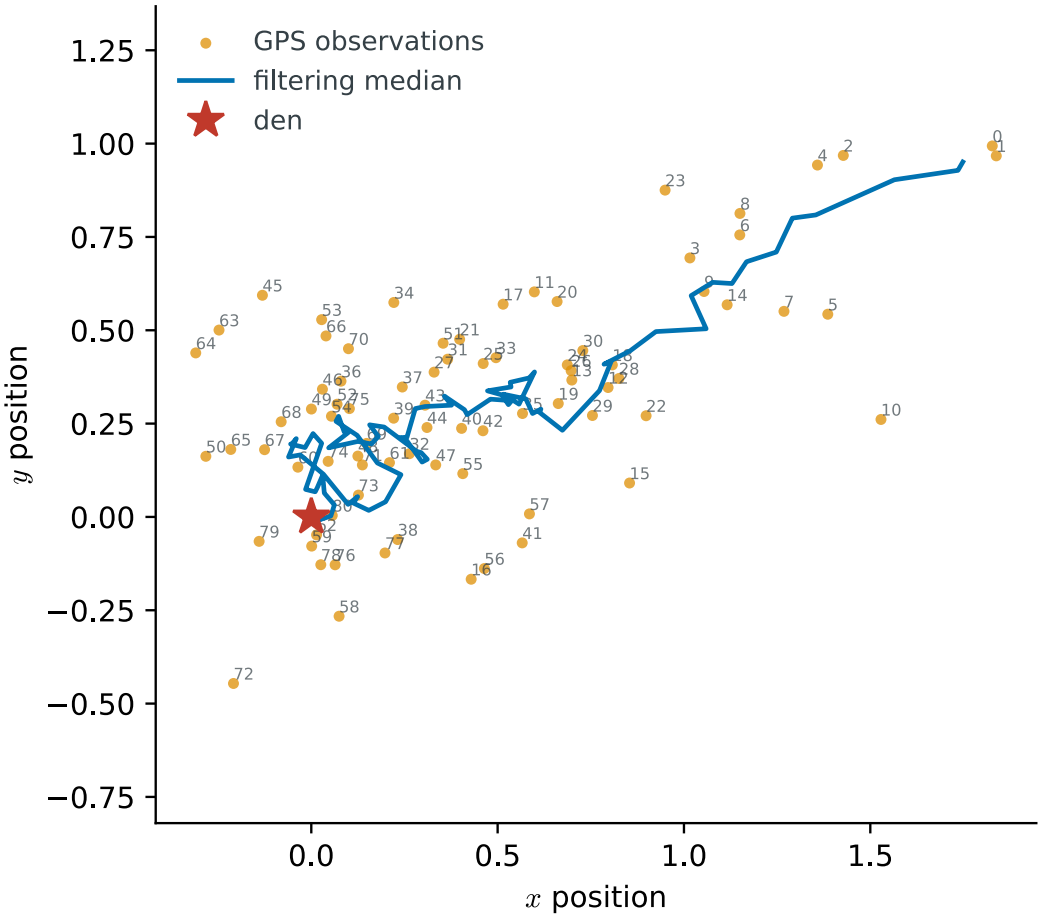
Therefore, the filtering distribution is given by the conditional distribution

$$p(\mathbf{z}_t|\mathbf{x}_{1:t-1}, \mathbf{x}_t) = p(\mathbf{z}_t|\mathbf{x}_{1:t}) = \mathcal{N}(\mathbf{z}_t|\mathbf{m}_t, \mathbf{P}_t),$$

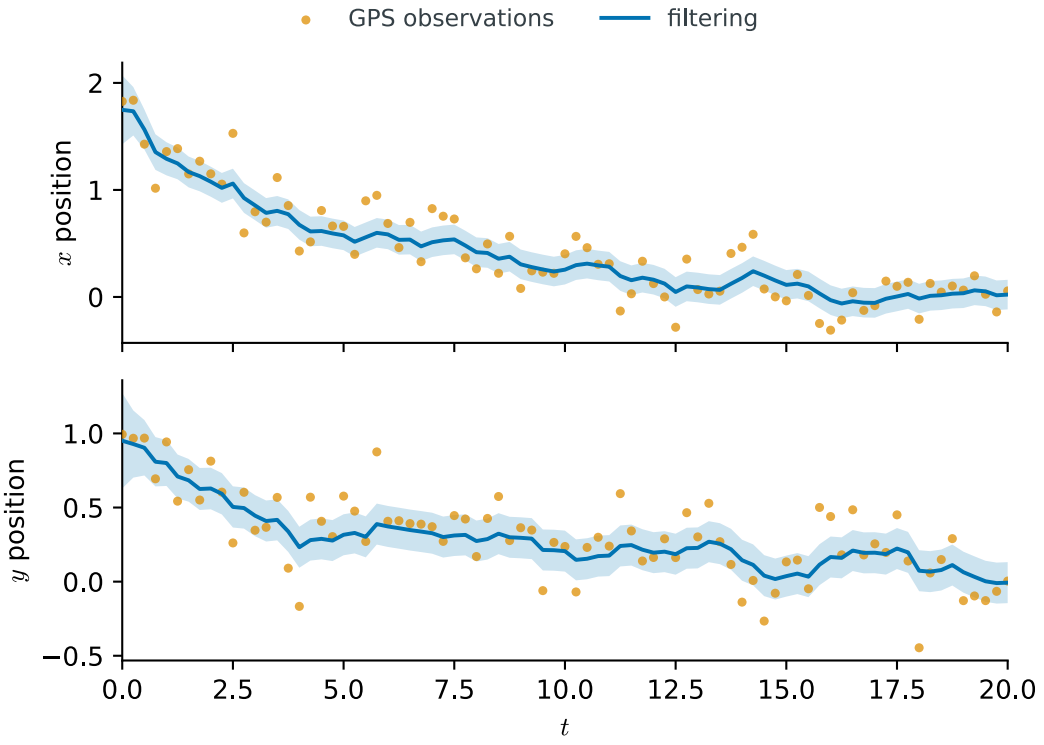
where

$$\mathbf{m}_t = \mathbf{m}_t^- + \mathbf{K}_t(\mathbf{x}_t - \mathbf{H}\mathbf{m}_t^-),$$
$$\mathbf{P}_t = (\mathbf{I} - \mathbf{K}_t\mathbf{H})\mathbf{P}_t^-,$$

and $\mathbf{K}_t = \mathbf{P}_t^-\mathbf{H}^T(\mathbf{H}\mathbf{P}_t^-\mathbf{H}^T + \mathbf{R})^{-1}$ is the Kalman gain and represents the weight given to the new observation.



The median of the filtering distribution at each time step.



The filtering distribution at each time step: median and 5-95% band.

The smoothing distributions $p(\mathbf{z}_t|\mathbf{x}_{1:T}) = \mathcal{N}(\mathbf{z}_t|\mathbf{m}_t^s, \mathbf{P}_t^s)$ are also Gaussian with mean $\mathbf{m}_t^s$ and covariance $\mathbf{P}_t^s$.

The parameters can be computed recursively using the **Rauch-Tung-Striebel smoother** equations,

$$\begin{aligned}
\mathbf{C}_t &= \mathbf{P}_t \mathbf{A}^T (\mathbf{P}_{t+1}^-)^{-1}, \\
\mathbf{m}_t^s &= \mathbf{m}_t + \mathbf{C}_t(\mathbf{m}_{t+1}^s - \mathbf{m}_{t+1}^-), \\
\mathbf{P}_t^s &= \mathbf{P}_t + \mathbf{C}_t(\mathbf{P}_{t+1}^s - \mathbf{P}_{t+1}^-)\mathbf{C}_t^T,
\end{aligned}$$

for $t = T-1, T-2, \ldots, 1$, with the base case $\mathbf{m}_T^s = \mathbf{m}_T$ and $\mathbf{P}_T^s = \mathbf{P}_T$.

Proof. Combining the filtering distribution $p(\mathbf{z}_t|\mathbf{x}_{1:t}) = \mathcal{N}(\mathbf{z}_t|\mathbf{m}_t, \mathbf{P}_t)$ with the transition model $p(\mathbf{z}_{t+1}|\mathbf{z}_t)$ gives a Gaussian joint,

$$p(\mathbf{z}_t, \mathbf{z}_{t+1}|\mathbf{x}_{1:t}) = p(\mathbf{z}_t|\mathbf{x}_{1:t})p(\mathbf{z}_{t+1}|\mathbf{z}_t) = \mathcal{N}\left(\begin{pmatrix}\mathbf{z}_t \\ \mathbf{z}_{t+1}\end{pmatrix} | \begin{bmatrix}\mathbf{m}_t \\ \mathbf{m}_{t+1}^-\end{bmatrix}, \begin{bmatrix}\mathbf{P}_t & \mathbf{P}_t\mathbf{A}^T \\ \mathbf{A}\mathbf{P}_t & \mathbf{P}_{t+1}^-\end{bmatrix}\right).$$

Given $\mathbf{z}_{t+1}$, the later observations $\mathbf{x}_{t+1:T}$ carry no further information on $\mathbf{z}_t$ (Markov property). Conditioning the joint on $\mathbf{z}_{t+1}$ gives

$$p(\mathbf{z}_t|\mathbf{z}_{t+1}, \mathbf{x}_{1:T}) = \mathcal{N}(\mathbf{z}_t|\mathbf{m}_t + \mathbf{C}_t(\mathbf{z}_{t+1} - \mathbf{m}_{t+1}^-), \mathbf{P}_t - \mathbf{C}_t\mathbf{P}_{t+1}^-\mathbf{C}_t^T).$$

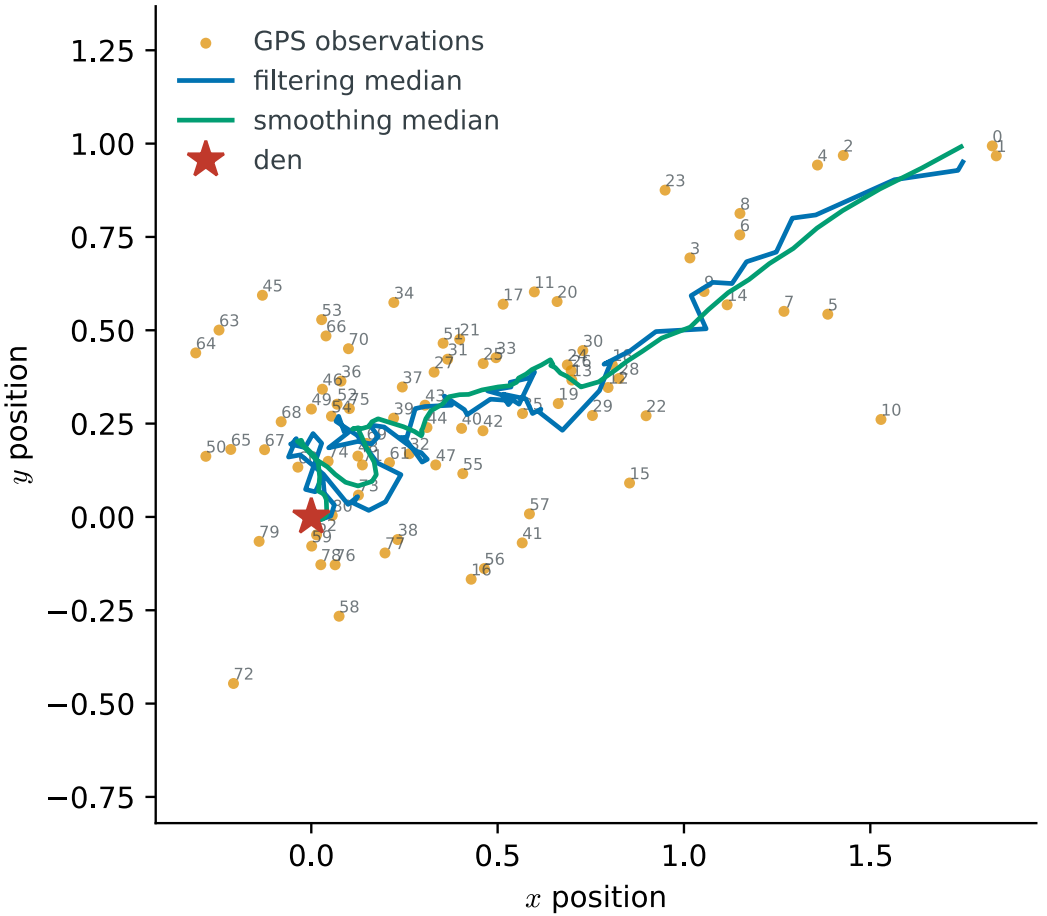
Multiplying by the smoothing distribution $p(\mathbf{z}_{t+1}|\mathbf{x}_{1:T}) = \mathcal{N}(\mathbf{z}_{t+1}|\mathbf{m}^s_{t+1}, \mathbf{P}^s_{t+1})$, computed at the previous step of the recursion, gives the joint

$$\begin{aligned} p(\mathbf{z}_{t+1}, \mathbf{z}_t|\mathbf{x}_{1:T}) &= p(\mathbf{z}_{t+1}|\mathbf{x}_{1:T}) p(\mathbf{z}_t|\mathbf{z}_{t+1}, \mathbf{x}_{1:T}) \\ &= \mathcal{N}\left(\begin{pmatrix} \mathbf{z}_{t+1} \\ \mathbf{z}_t \end{pmatrix} | \begin{bmatrix} \mathbf{m}^s_{t+1} \\ \mathbf{m}^s_t \end{bmatrix}, \begin{bmatrix} \mathbf{P}^s_{t+1} & \mathbf{P}^s_{t+1}\mathbf{C}_t^T \\ \mathbf{C}_t\mathbf{P}^s_{t+1} & \mathbf{P}^s_t \end{bmatrix} \right), \end{aligned}$$

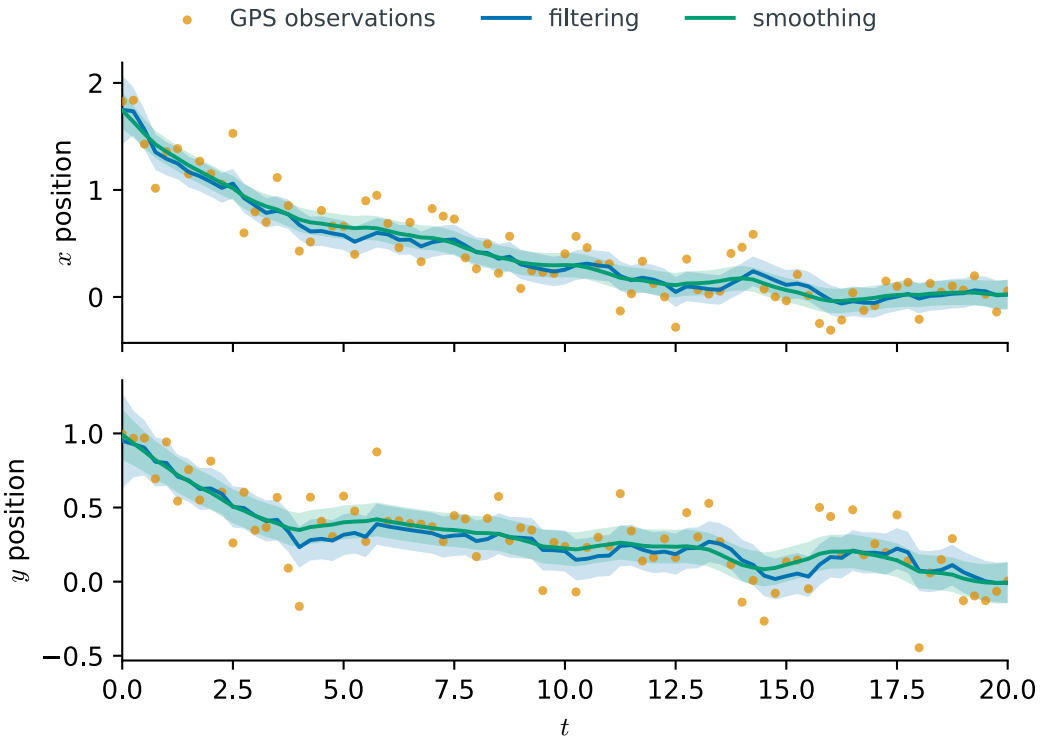
where

$$\begin{aligned} \mathbf{m}^s_t &= \mathbf{m}_t + \mathbf{C}_t(\mathbf{m}^s_{t+1} - \mathbf{m}^-_{t+1}), \\ \mathbf{P}^s_t &= \mathbf{P}_t - \mathbf{C}_t\mathbf{P}^-_{t+1}\mathbf{C}_t^T + \mathbf{C}_t\mathbf{P}^s_{t+1}\mathbf{C}_t^T = \mathbf{P}_t + \mathbf{C}_t(\mathbf{P}^s_{t+1} - \mathbf{P}^-_{t+1})\mathbf{C}_t^T. \end{aligned}$$

Its marginal over $\mathbf{z}_t$ is the smoothing distribution $\mathcal{N}(\mathbf{z}_t|\mathbf{m}^s_t, \mathbf{P}^s_t)$.



The median of the smoothing distribution at each time step.



The smoothing distribution at each time step: median and 5-95% band.

Hidden Markov models

A **hidden Markov model** (HMM) is a state-space model whose states are discrete, $z_t \in \{1, \dots, K\}$, with transition probabilities

$$p(z_t = j \mid z_{t-1} = i) = \mathbf{A}_{i,j},$$

and any observation model $p(x_t \mid z_t = j)$ for each state j. It is a mixture model whose component switches over time.

Example. The behavior of a wolf (resting, foraging, traveling) as the state, its speed as the observation. $\mathbf{A}$ says how often the wolf switches from one behavior to another.

With discrete states, the integrals of the Bayes filter and smoother become sums over the K states.

1. Forward pass, the Bayes filter. For $t = 1, \dots, T$,

$$p(z_t = j \mid x_{1:t}) \propto p(x_t \mid z_t = j) \sum_{i=1}^{K} \mathbf{A}_{i,j}\, p(z_{t-1} = i \mid x_{1:t-1}),$$

   where the sum is the prior $p(z_1 = j)$ at $t = 1$.

2. Backward pass, the Bayes smoother. For $t = T-1, \dots, 1$,

$$p(z_t = i \mid x_{1:T}) = p(z_t = i \mid x_{1:t}) \sum_{j=1}^{K} \frac{\mathbf{A}_{i,j}\, p(z_{t+1} = j \mid x_{1:T})}{p(z_{t+1} = j \mid x_{1:t})},$$

   where $p(z_{t+1} = j \mid x_{1:t}) = \sum_i \mathbf{A}_{i,j}\, p(z_t = i \mid x_{1:t})$.

Together, the two passes form the **forward-backward algorithm**. Each costs $K^2 T$ operations.

Example. A wolf switches between $K = 3$ behaviors, resting, foraging and traveling, and we observe its speed x_t.

- The transition matrix keeps each behavior for a while,

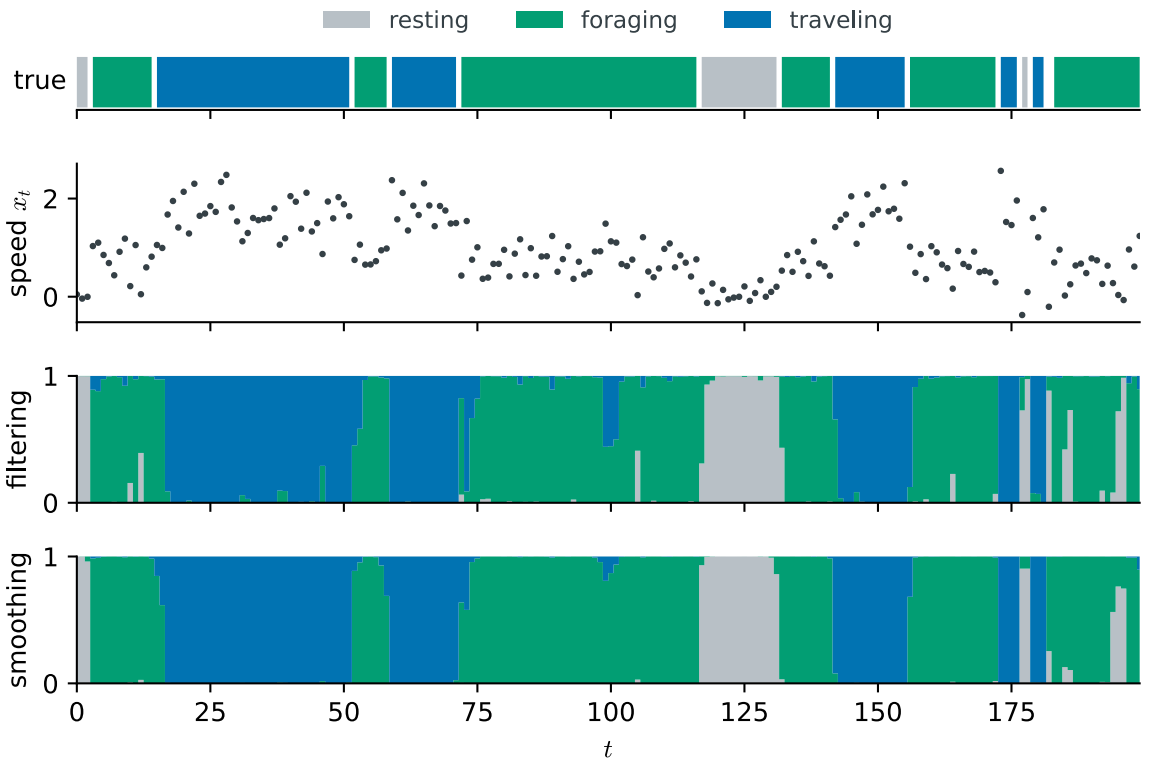
$$\mathbf{A} = \begin{bmatrix} 0.95 & 0.04 & 0.01 \\ 0.03 & 0.93 & 0.04 \\ 0.02 & 0.06 & 0.92 \end{bmatrix}.$$

- The speed is Gaussian given the behavior,

$$p(x_t \mid z_t = j) = \mathcal{N}(x_t \mid \mu_j, \sigma_j^2), \quad \boldsymbol{\mu} = (0.1, 0.7, 1.6), \quad \boldsymbol{\sigma} = (0.15, 0.3, 0.4).$$

- The wolf starts at rest, $p(z_1 = 1) = 1$.

We simulate $T = 200$ steps from this model, then run the forward-backward algorithm on the speeds alone.



A simulated wolf switching between three behaviors, observed through its speed.
Its most probable state is right 94% of the time filtering, 97% smoothing.

Learning the parameters

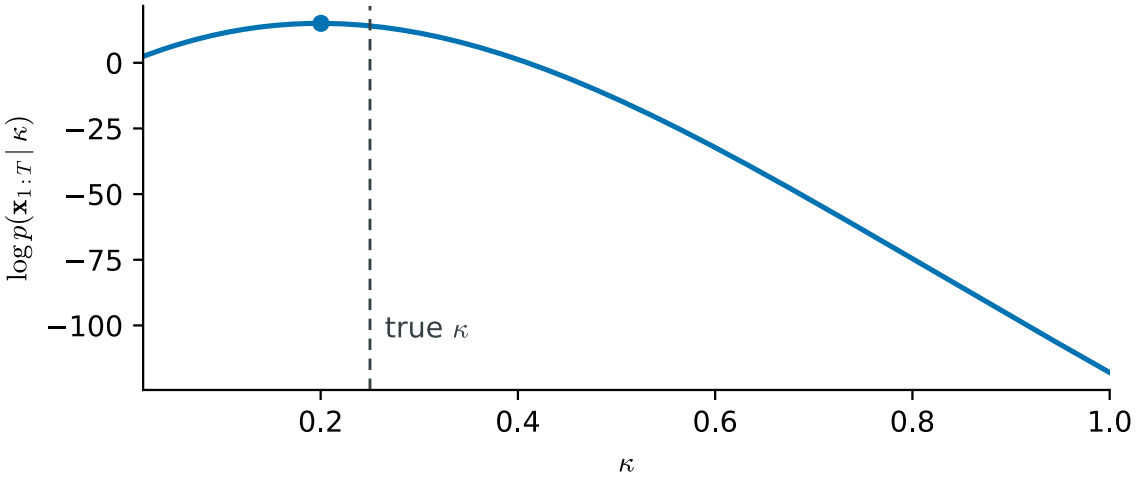
So far, the parameters $\theta = (\kappa, \sigma, \mathbf{R}, \boldsymbol{\mu})$ of the wolf model were taken as known. The Bayes filter also gives their likelihood. At each step, the update normalizes by

$$p(\mathbf{x}_t \mid \mathbf{x}_{1:t-1}, \theta) = \int p(\mathbf{x}_t \mid \mathbf{z}_t, \theta)\, p(\mathbf{z}_t \mid \mathbf{x}_{1:t-1}, \theta)\, d\mathbf{z}_t.$$

The product of these normalizers is the marginal likelihood of the data,

$$p(\mathbf{x}_{1:T} \mid \theta) = \prod_{t=1}^{T} p(\mathbf{x}_t \mid \mathbf{x}_{1:t-1}, \theta).$$

It can be maximized over θ, or combined with a prior $p(\theta)$ into a posterior.



$$\log p(\mathbf{x}_{1:T} \mid \kappa) = \sum_{t=1}^{T} \log p(\mathbf{x}_t \mid \mathbf{x}_{1:t-1}, \kappa)$$

The log-likelihood of the GPS observations, as a function of κ.
It peaks at 0.20; the data were simulated with $\kappa = 0.25$.

Continuous-time models



So far, time advanced in fixed steps Δt, at which the transition and observation models were defined.

We now model the state in **continuous time**. Physical processes are naturally described by rates of change. Observations often arrive at irregular times, triggered by events or collected at several time scales.

From discrete to continuous time

In discrete-time state-space models, we considered transition models of the form $p(\mathbf{z}_t|\mathbf{z}_{t-1})$ that describe how the state evolves from one time step to the next. For additive transitions and additive noise, this can be expressed as

$$\mathbf{z}_t = \mathbf{z}_{t-1} + f(\mathbf{z}_{t-1})\Delta t + \mathbf{w}_t,$$

where f is a deterministic function and $\mathbf{w}_t$ is random noise.

Shuffling the terms, we get

$$\frac{\mathbf{z}_t - \mathbf{z}_{t-1}}{\Delta t} = f(\mathbf{z}_{t-1}) + \frac{\mathbf{w}_t}{\Delta t},$$

which, in the limit as $\Delta t \to 0$, gives us a continuous-time model

$$\frac{d\mathbf{z}(t)}{dt} = f(\mathbf{z}(t)) + \frac{d\mathbf{w}(t)}{dt},$$

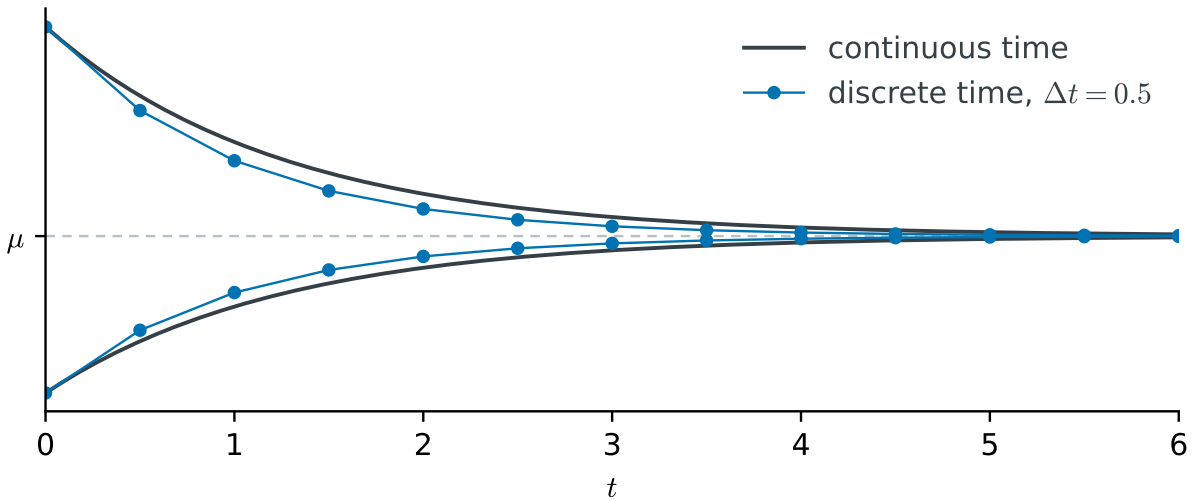
where $\mathbf{z}(t)$ is the state at time t and $\mathbf{w}(t)$ is continuous-time noise.

Without the noise, we get a **deterministic** system, an **ordinary differential equation** (ODE),

$$\frac{d\mathbf{z}(t)}{dt} = f(\mathbf{z}(t)).$$

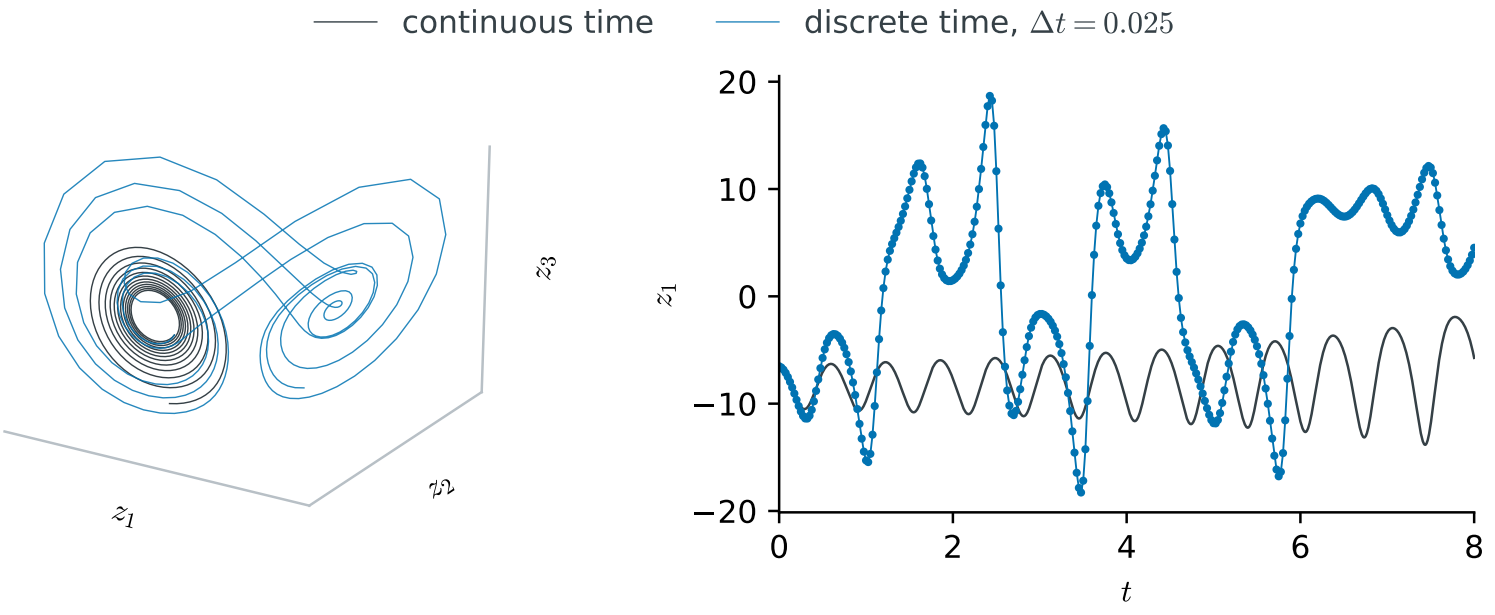
Example 1. Exponential decay to an equilibrium point μ,

$$\frac{dz(t)}{dt} = -\kappa(z(t) - \mu), \quad \text{with solution} \quad z(t) = \mu + (z(0) - \mu)e^{-\kappa t}.$$



Example 2. The **Lorenz system** is a chaotic ODE in three variables,

$$\frac{dz_1}{dt} = s(z_2 - z_1), \quad \frac{dz_2}{dt} = z_1(r - z_3) - z_2, \quad \frac{dz_3}{dt} = z_1 z_2 - b z_3.$$



The two trajectories part after about one time unit.

Brownian motion

To add stochasticity to the ODE, we need a continuous-time stochastic process that can model random noise.

We can model the noise term $\mathbf{w}(t)$ as a standard Brownian motion (Wiener process) $W(t)$, which has the following properties:

- $W(0) = 0$.
- For $0 \leq s < t$, the increment $W(t) - W(s) \sim \mathcal{N}(0, t - s)$.
- Increments over disjoint intervals are independent.
- $W(t)$ is continuous in t.

A vector $\mathbf{W}(t)$ has independent Brownian components.

Adding Brownian motion to the ODE, scaled by a **diffusion term** g, gives a **stochastic differential equation** (SDE),

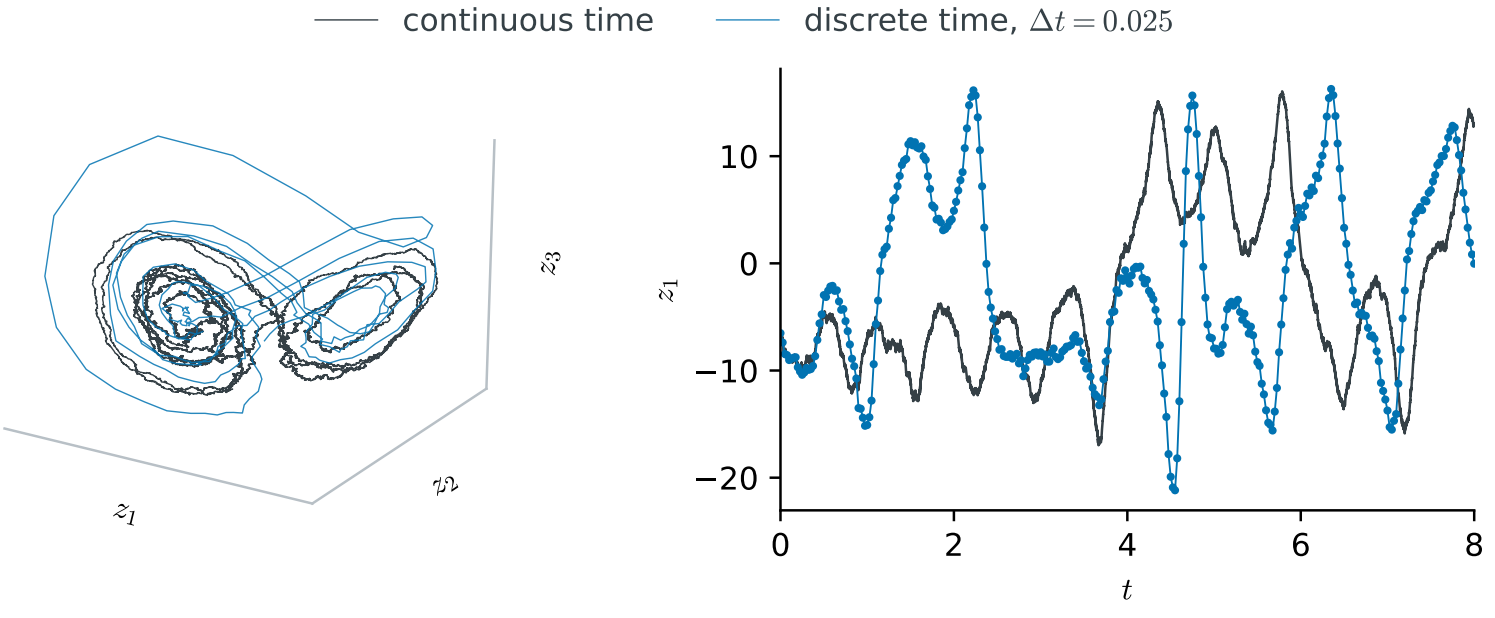
$$d\mathbf{z}(t) = f(\mathbf{z}(t))\, dt + g(\mathbf{z}(t))\, d\mathbf{W}(t).$$

Over an infinitesimal interval dt, the drift $f\, dt$ is the deterministic change of the state, and the diffusion $g\, d\mathbf{W}$ its random change.

Brownian motion is nowhere differentiable, so the white noise $d\mathbf{W}(t)/dt$ is only symbolic. The differential form avoids it.

Example 2, with noise. The Lorenz system becomes the SDE

$$d\mathbf{z} = f(\mathbf{z})\,dt + \sigma\,d\mathbf{W}.$$



In continuous and in discrete time, driven by the same Brownian motion.

Example 3. The wolf model in continuous time is Example 1 in two dimensions, towards the home location $\boldsymbol{\mu}$, with noise,

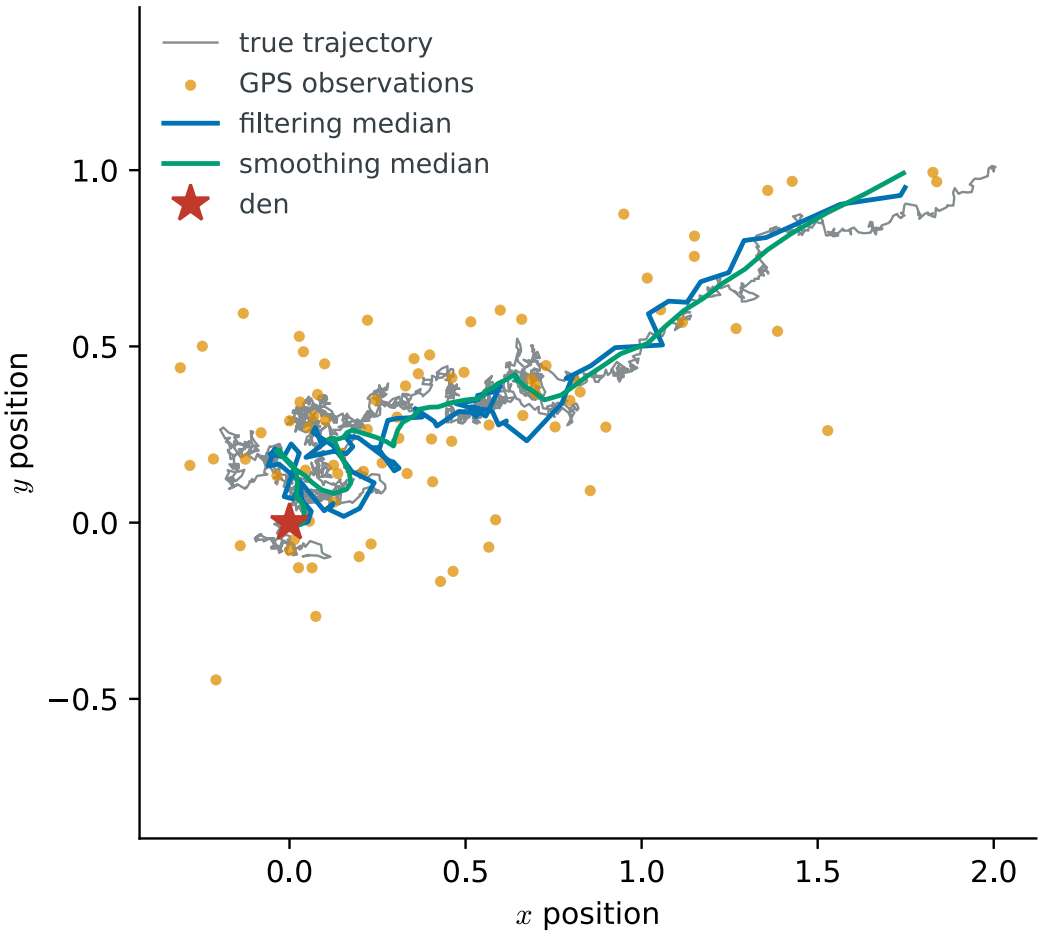
$$d\mathbf{z}(t) = -\kappa(\mathbf{z}(t) - \boldsymbol{\mu})\, dt + \sigma\, d\mathbf{W}(t),$$

where $\kappa > 0$ is the strength of the attraction and σ the diffusion coefficient. This SDE is the **Ornstein-Uhlenbeck process**.

The discrete-time model of Part I,

$$\mathbf{z}_t = \mathbf{z}_{t-1} - \kappa(\mathbf{z}_{t-1} - \boldsymbol{\mu})\Delta t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 \Delta t\, \mathbf{I}),$$

is its Euler-Maruyama discretization with step Δt.



The true trajectory behind the GPS observations: an Ornstein-Uhlenbeck process, simulated on a fine grid and observed every 0.25 time units.

The OU process can be discretized exactly, even at irregular observation times $t_1 < t_2 < \dots$. With $\Delta_i = t_i - t_{i-1}$,

$$p(\mathbf{z}(t_i) \mid \mathbf{z}(t_{i-1})) = \mathcal{N}\left(\boldsymbol{\mu} + e^{-\kappa\Delta_i}(\mathbf{z}(t_{i-1}) - \boldsymbol{\mu}),\ \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa\Delta_i}\right)\mathbf{I}\right).$$

This is a linear Gaussian transition, with $\mathbf{A}_i = e^{-\kappa\Delta_i}\mathbf{I}$ and $\mathbf{b}_i = (1 - e^{-\kappa\Delta_i})\boldsymbol{\mu}$. The Kalman filter and smoother apply unchanged, with matrices that change with the time step. The figures of this lecture use it, at $\Delta = 0.25$.

Continuous or discrete time?

Continuous time

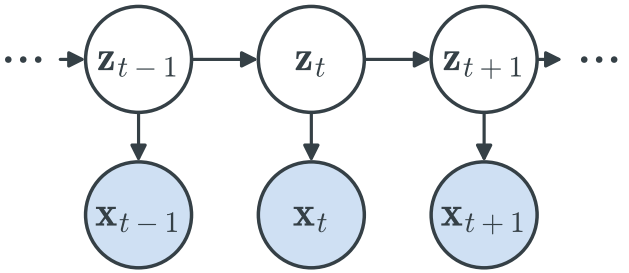
- irregular observations
- physical laws, rates of change
- parameters are rates, such as κ

Discrete time

- regular sampling
- no mechanistic model
- simpler to compute

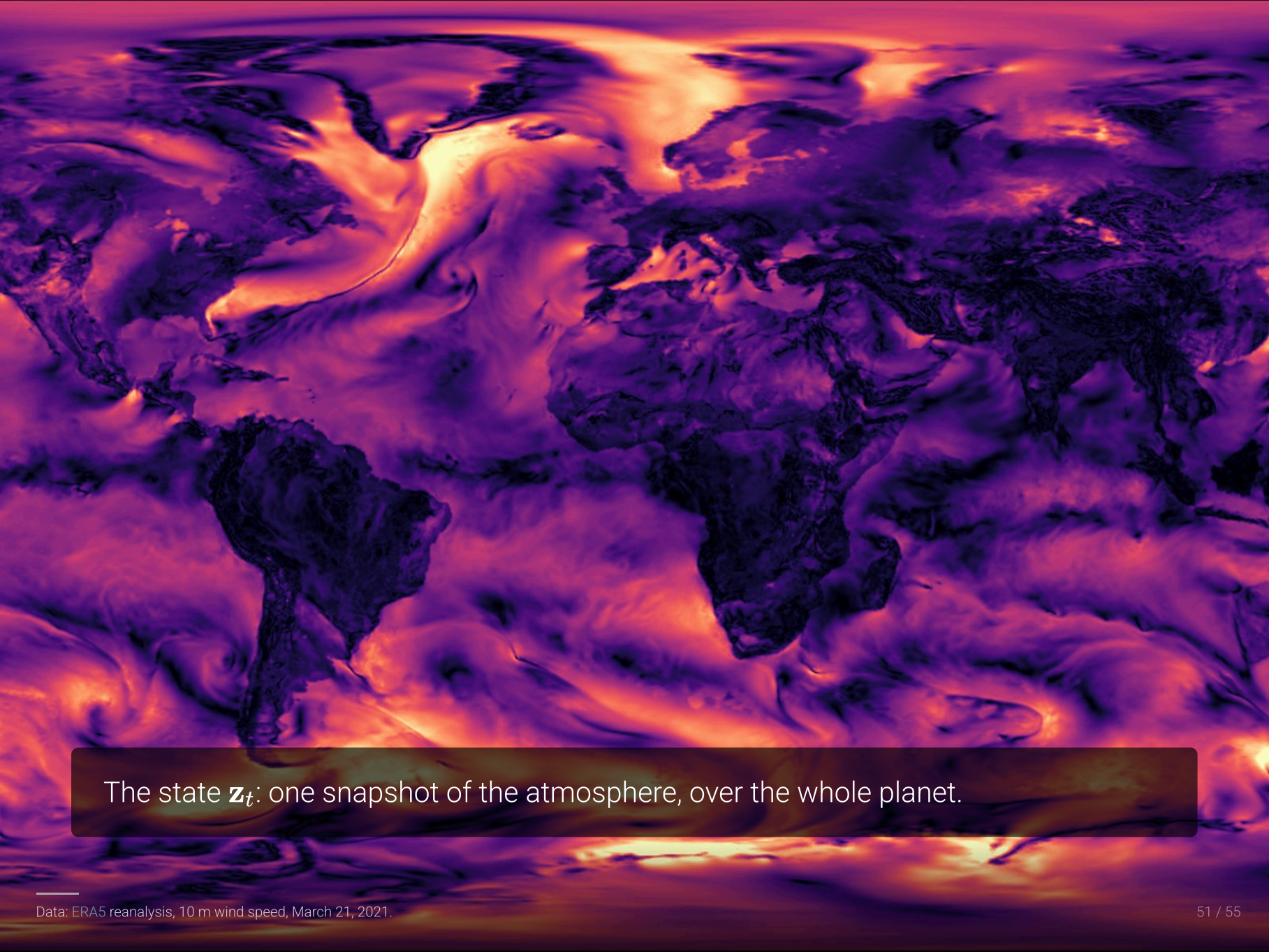
> Model in continuous time, discretize for computation, exactly when possible.

Application: weather forecasting

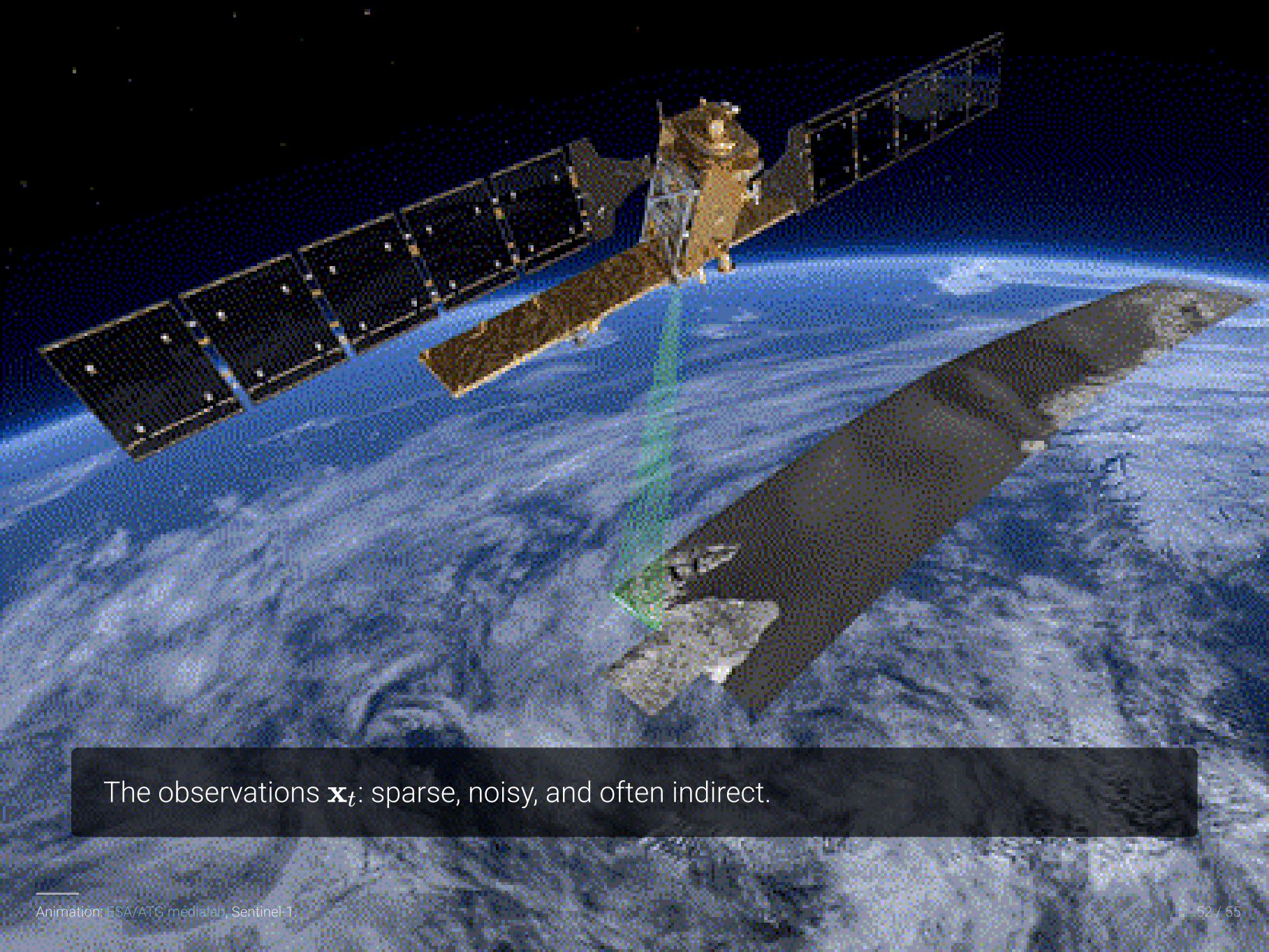


Weather forecasting is a state-space model:

- the state $\mathbf{z}_t$ is the atmosphere on a three-dimensional grid (temperature, wind, pressure, humidity), about 10^9 numbers;
- the transition model $p(\mathbf{z}_t|\mathbf{z}_{t-1})$ is the physics of the atmosphere, discretized in time with steps of a few minutes;
- the observations $\mathbf{x}_t$ come from satellites, weather stations, balloons and aircraft, tens of millions every day, through an observation model $p(\mathbf{x}_t|\mathbf{z}_t)$.

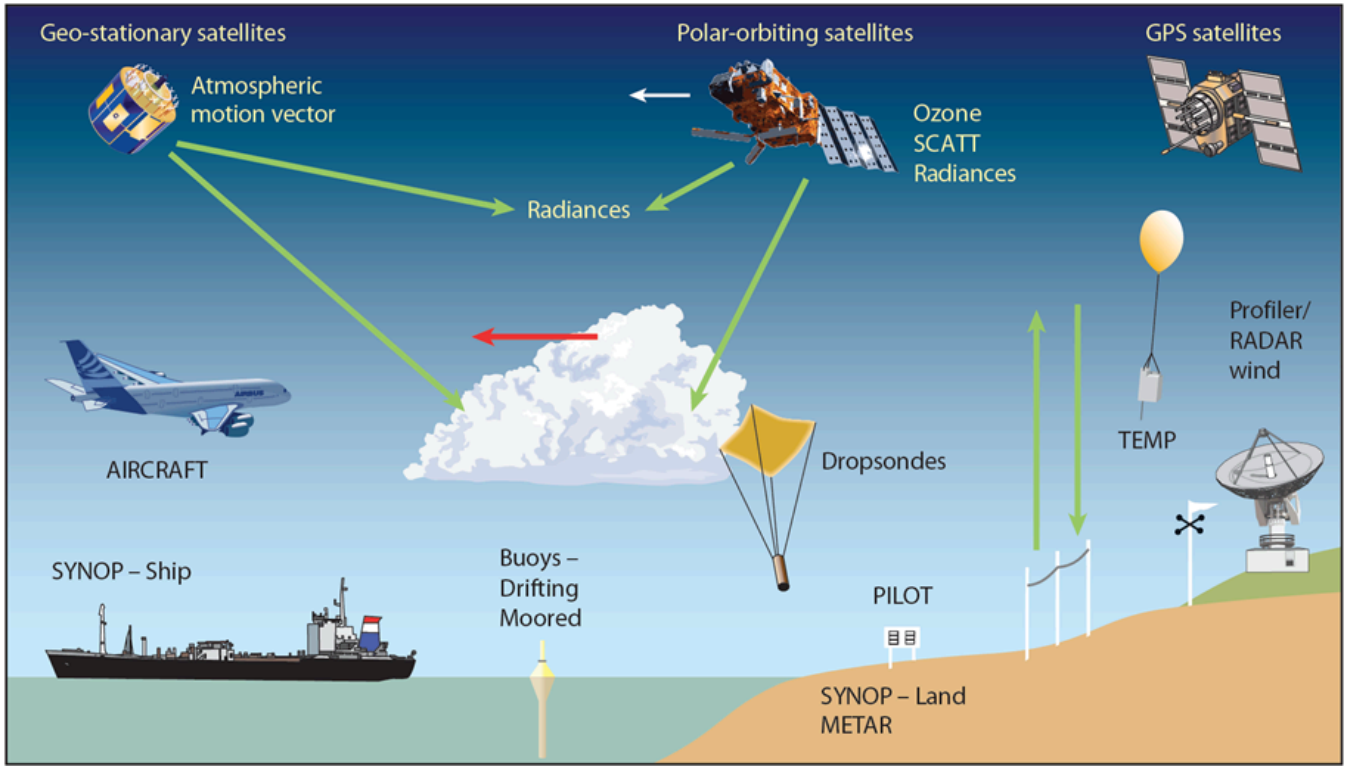


Data: ERA5 reanalysis, 10 m wind speed, March 21, 2021.



The observations $\mathbf{x}_t$: sparse, noisy, and often indirect.

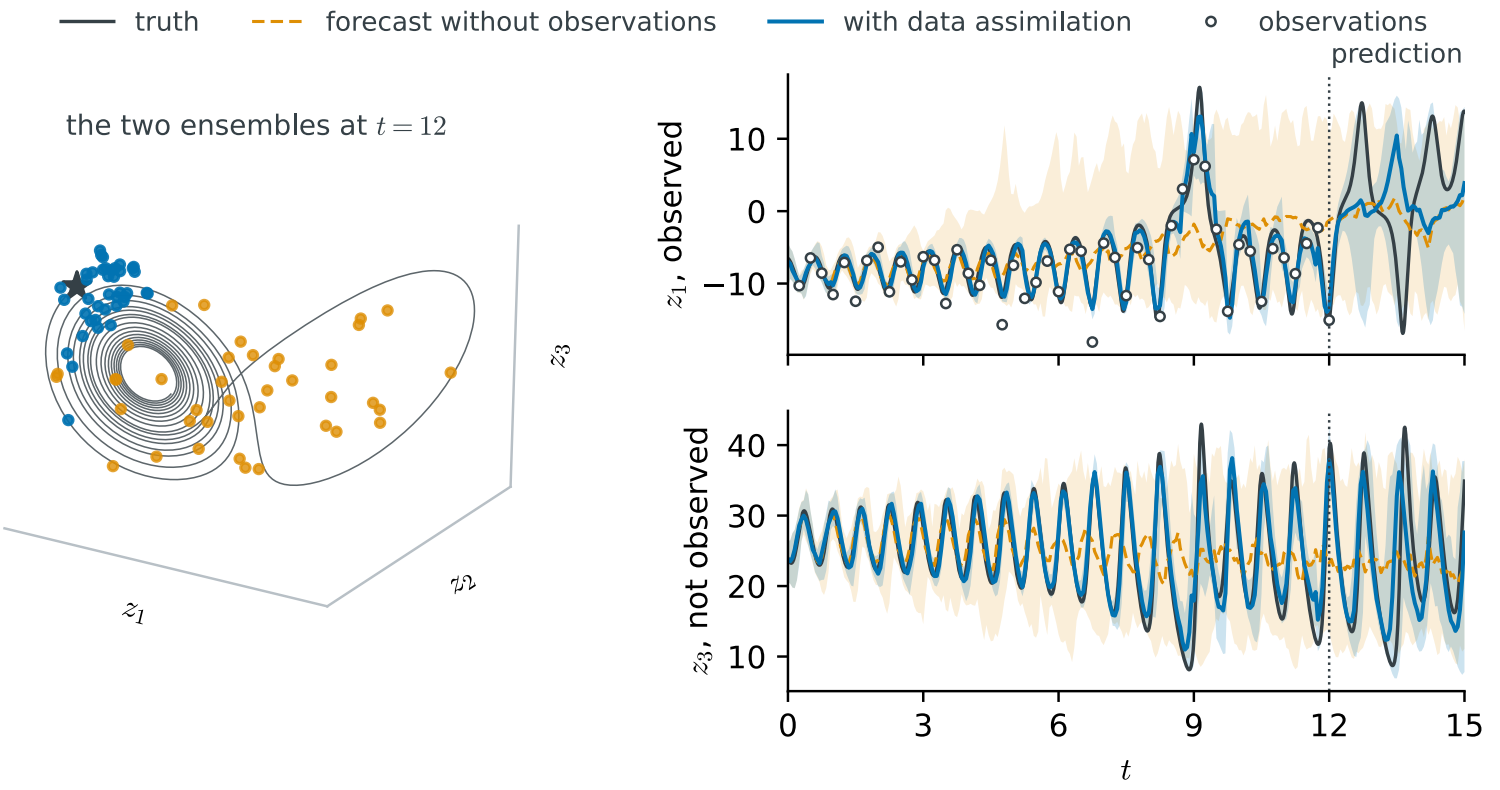
Animation: ESA/ATG medialab, Sentinel-1.



The observations assimilated every day in the ECMWF forecasting system.

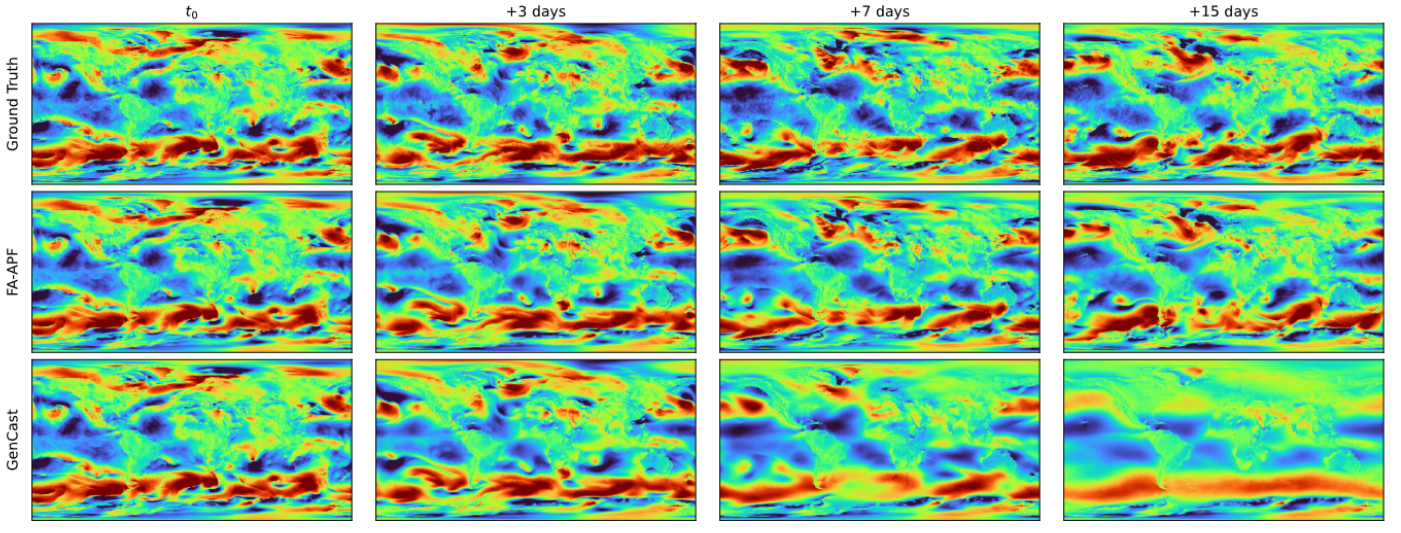
Credits: ECMWF, CC BY 4.0.

Forecasting centres run the Bayes filter continuously: every 12 hours, they update the forecast with the latest observations, then predict. This cycle is called **data assimilation**. On the Lorenz system, with only z_1 observed:



With data assimilation, the forecast tracks z_3, then predicts after $t = 12$.
Without observations, it spreads over the attractor.

On the real atmosphere, the transition model can be a machine-learning forecaster such as GenCast, and the filter an ensemble of states updated with each new observation (a particle filter):



10 m zonal wind: truth (ERA5), filter with observations, forecast alone.
Without observations, the forecast drifts away within a week.

Credits: Savary et al, ICML 2026.

The end.